

DSC 291: Learning Theory Assignment 8

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1 Part A: Choosing Regularization by Validation

1.1 Problem 1. Validation Selection for a Finite Candidate Set

Condition on fixed predictors

$$h_1, \dots, h_K$$

that are independent of the validation set V . Define

$$\Delta_k := L_{\mathcal{D}}(h_k) - L_V(h_k).$$

Let

$$k^* \in \arg \min_{k \in [K]} L_{\mathcal{D}}(h_k), \quad \hat{k} \in \arg \min_{k \in [K]} L_V(h_k).$$

Then

$$L_{\mathcal{D}}(h_{\hat{k}}) = L_V(h_{\hat{k}}) + \Delta_{\hat{k}} \leq L_V(h_{k^*}) + \Delta_{\hat{k}} = L_{\mathcal{D}}(h_{k^*}) - \Delta_{k^*} + \Delta_{\hat{k}}.$$

Therefore

$$L_{\mathcal{D}}(h_{\hat{k}}) \leq \min_{k \in [K]} L_{\mathcal{D}}(h_k) + 2 \max_{k \in [K]} |\Delta_k|.$$

It remains to bound the expected maximum validation deviation.

For fixed k , $L_V(h_k)$ is the average of n_V independent $[0, 1]$ losses with mean $L_{\mathcal{D}}(h_k)$. Hoeffding's lemma gives, for every $s > 0$,

$$\mathbb{E}_V e^{s\Delta_k} \leq e^{s^2/(8n_V)}, \quad \mathbb{E}_V e^{-s\Delta_k} \leq e^{s^2/(8n_V)}.$$

Hence

$$\mathbb{E}_V e^{s|\Delta_k|} \leq \mathbb{E}_V e^{s\Delta_k} + \mathbb{E}_V e^{-s\Delta_k} \leq 2e^{s^2/(8n_V)}.$$

Using the exponential maximum bound,

$$\mathbb{E}_V \max_k |\Delta_k| \leq \frac{1}{s} \log \left(\sum_{k=1}^K \mathbb{E}_V e^{s|\Delta_k|} \right) \leq \frac{\log(2K)}{s} + \frac{s}{8n_V}.$$

Optimizing at

$$s = \sqrt{8n_V \log(2K)}$$

gives

$$\mathbb{E}_V \max_k |\Delta_k| \leq \sqrt{\frac{\log(2K)}{2n_V}}.$$

Taking expectation in the earlier inequality yields

$$\mathbb{E}_V L_{\mathcal{D}}(h_{\hat{k}}) \leq \min_{k \in [K]} L_{\mathcal{D}}(h_k) + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

1.2 Problem 2. Oracle Inequality over a Grid

Condition on the training sample T . Then the trained predictors

$$\{h_\lambda : \lambda \in \Lambda\}$$

are fixed with respect to the validation set V , so Problem 1 gives

$$\mathbb{E}_V [L_{\mathcal{D}}(h_{\hat{\lambda}}) \mid T] \leq \min_{\lambda \in \Lambda} L_{\mathcal{D}}(h_\lambda) + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

Now take expectation over T :

$$\mathbb{E} L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq \min_{\lambda \in \Lambda} \mathbb{E}_T L_{\mathcal{D}}(A_\lambda(T)) + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

For each fixed λ , the Week 8 stable RERM theorem gives, for every $u \in \mathcal{W}$,

$$\mathbb{E}_T L_{\mathcal{D}}(A_\lambda(T)) \leq L_{\mathcal{D}}(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T}.$$

Taking the infimum over u and then the minimum over $\lambda \in \Lambda$ gives

$$\mathbb{E} L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq \min_{\lambda \in \Lambda} \inf_{u \in \mathcal{W}} \left\{ L_{\mathcal{D}}(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T} \right\} + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

The validation term is only logarithmic in K because the validation set is used to choose among K already-trained candidates. Uniformly controlling the K validation estimates costs

$$O\left(\sqrt{\frac{\log K}{n_V}}\right),$$

not $O(\sqrt{K/n_V})$.

1.3 Problem 3. Adapting to an Unknown Comparator Scale

Fix a comparator u and write

$$a := \Psi(u).$$

The λ -dependent part of the bound from Problem 2 is

$$f(\lambda) := \lambda a + \frac{2G^2}{\lambda \alpha n_T}.$$

Set

$$c := \frac{2G^2}{\alpha n_T}.$$

Then

$$f(\lambda) = a\lambda + \frac{c}{\lambda}.$$

The continuous optimizer is

$$\lambda_{\text{opt}}(a) = \sqrt{\frac{c}{a}} = \sqrt{\frac{2G^2}{\alpha a n_T}},$$

and the optimal value is

$$f(\lambda_{\text{opt}}) = 2\sqrt{ac} = 2G\sqrt{\frac{2a}{\alpha n_T}}.$$

By the grid assumption, choose $\lambda \in \Lambda$ with

$$\frac{\lambda_{\text{opt}}}{2} \leq \lambda \leq 2\lambda_{\text{opt}}.$$

Write $\lambda = t\lambda_{\text{opt}}$ with $t \in [1/2, 2]$. Since $a\lambda_{\text{opt}} = c/\lambda_{\text{opt}} = \sqrt{ac}$,

$$f(\lambda) = \sqrt{ac} \left(t + \frac{1}{t} \right).$$

On $[1/2, 2]$,

$$t + \frac{1}{t} \leq \frac{5}{2}.$$

Therefore

$$f(\lambda) \leq \frac{5}{2}\sqrt{ac} = \frac{5\sqrt{2}}{2} G \sqrt{\frac{a}{\alpha n_T}}.$$

Plugging this choice into the oracle inequality gives, for every u satisfying

$$B_{\min}^2 \leq \Psi(u) \leq B_{\max}^2,$$

the bound

$$\mathbb{E}L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq L_{\mathcal{D}}(u) + \frac{5\sqrt{2}}{2} G \sqrt{\frac{\Psi(u)}{\alpha n_T}} + 2\sqrt{\frac{\log(2K)}{2n_V}}.$$

1.4 Problem 4. A Dyadic Grid and the Price of Tuning

For

$$a \in [B_{\min}^2, B_{\max}^2],$$

the optimal regularization parameter ranges over

$$\lambda_{\text{opt}}(a) \in \left[\frac{\sqrt{2}G}{B_{\max}\sqrt{\alpha n_T}}, \frac{\sqrt{2}G}{B_{\min}\sqrt{\alpha n_T}} \right].$$

Define

$$\lambda_0 := \frac{\sqrt{2}G}{B_{\max}\sqrt{\alpha n_T}}$$

and use the dyadic grid

$$\Lambda = \{\lambda_0 2^j : j = 0, 1, \dots, J\}, \quad J := \left\lceil \log_2 \left(\frac{B_{\max}}{B_{\min}} \right) \right\rceil.$$

Then

$$K = J + 1 = \left\lceil \log_2 \left(\frac{B_{\max}}{B_{\min}} \right) \right\rceil + 1.$$

For every λ_{opt} in the above interval, the first grid point at least as large as λ_{opt} is at most $2\lambda_{\text{opt}}$, and is certainly at least $\lambda_{\text{opt}}/2$. Thus the grid satisfies the required property.

Now assume n is even and set

$$n_T = n_V = \frac{n}{2}.$$

The result from Problem 3 becomes

$$\mathbb{E}L_{\mathcal{D}}(h_{\hat{\lambda}}) \leq L_{\mathcal{D}}(u) + 5G \sqrt{\frac{\Psi(u)}{\alpha n}} + 2\sqrt{\frac{\log(2K)}{n}}.$$

The first statistical term,

$$5G \sqrt{\frac{\Psi(u)}{\alpha n}},$$

is the RERM learning term. The second term,

$$2\sqrt{\frac{\log(2K)}{n}},$$

is the validation-selection term.

If the comparator scale $\Psi(u)$ were known in advance, one could tune λ directly to that scale and avoid selecting over a grid. Not knowing the scale introduces the validation term involving

$$\log K = \log(\lceil \log_2(B_{\max}/B_{\min}) \rceil + 1),$$

and also a small constant-factor loss from discretizing λ .

2 Part B: Approximate RERM and Optimization Error

2.1 Problem 1. Approximate Minimizers are Close to Exact Minimizers

For a fixed sample S , the objective

$$F_S(w) = L_S(w) + \lambda\Psi(w)$$

is $\lambda\alpha$ -strongly convex because L_S is convex and Ψ is α -strongly convex. Since w_S is its exact minimizer,

$$F_S(w) \geq F_S(w_S) + \frac{\lambda\alpha}{2}\|w - w_S\|^2$$

for every $w \in \mathcal{W}$. Apply this with $w = \tilde{w}_S$:

$$F_S(\tilde{w}_S) \geq F_S(w_S) + \frac{\lambda\alpha}{2}\|\tilde{w}_S - w_S\|^2.$$

The optimization guarantee gives

$$F_S(\tilde{w}_S) \leq F_S(w_S) + \eta.$$

Combining the two inequalities,

$$\frac{\lambda\alpha}{2}\|\tilde{w}_S - w_S\|^2 \leq \eta.$$

Hence

$$\|\tilde{w}_S - w_S\| \leq \sqrt{\frac{2\eta}{\lambda\alpha}}.$$

2.2 Problem 2. Stability of Approximate RERM

Let S and S' differ in one example. For any test point z ,

$$|\ell(\tilde{w}_S, z) - \ell(\tilde{w}_{S'}, z)| \leq G\|\tilde{w}_S - \tilde{w}_{S'}\|.$$

By the triangle inequality,

$$\|\tilde{w}_S - \tilde{w}_{S'}\| \leq \|\tilde{w}_S - w_S\| + \|w_S - w_{S'}\| + \|w_{S'} - \tilde{w}_{S'}\|.$$

Problem 1 bounds the first and third terms, and the Week 8 exact movement bound gives

$$\|w_S - w_{S'}\| \leq \frac{2G}{\lambda\alpha n}.$$

Therefore

$$\|\tilde{w}_S - \tilde{w}_{S'}\| \leq 2\sqrt{\frac{2\eta}{\lambda\alpha}} + \frac{2G}{\lambda\alpha n}.$$

Multiplying by G yields

$$|\ell(\tilde{w}_S, z) - \ell(\tilde{w}_{S'}, z)| \leq \frac{2G^2}{\lambda\alpha n} + 2G\sqrt{\frac{2\eta}{\lambda\alpha}}.$$

2.3 Problem 3. Learning Guarantee with Optimization Error

Let w_S denote the exact RERM solution and $\tilde{w}_S$ the approximate solution. By Lipschitzness,

$$L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(w_S) + G\|\tilde{w}_S - w_S\|.$$

Using Problem 1,

$$L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(w_S) + G\sqrt{\frac{2\eta}{\lambda\alpha}}.$$

Taking expectation over S and applying the exact stable-RERM learning theorem to w_S gives, for every $u \in \mathcal{W}$,

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda\alpha n} + G\sqrt{\frac{2\eta}{\lambda\alpha}}.$$

2.4 Problem 4. How Accurate Must the Optimizer Be?

Assume

$$\Psi(u) \leq B^2.$$

Use

$$\lambda = \sqrt{\frac{2G^2}{\alpha B^2 n}} = \frac{\sqrt{2}G}{B\sqrt{\alpha n}}.$$

Then

$$\lambda B^2 = \sqrt{2}GB \frac{1}{\sqrt{\alpha n}},$$

and

$$\frac{2G^2}{\lambda \alpha n} = \sqrt{2}GB \frac{1}{\sqrt{\alpha n}}.$$

So the bound from Problem 3 becomes

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + 2\sqrt{2}GB \frac{1}{\sqrt{\alpha n}} + G\sqrt{\frac{2\eta}{\lambda \alpha}}.$$

We want the optimization term to satisfy

$$G\sqrt{\frac{2\eta}{\lambda \alpha}} \leq \frac{GB}{\sqrt{\alpha n}}.$$

After canceling G and squaring,

$$\frac{2\eta}{\lambda \alpha} \leq \frac{B^2}{\alpha n}.$$

Thus it is sufficient that

$$\eta \leq \frac{\lambda B^2}{2n} = \frac{GB}{\sqrt{2\alpha} n^{3/2}}.$$

Under this condition,

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + (2\sqrt{2} + 1) \frac{GB}{\sqrt{\alpha n}}.$$

Equivalently,

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq \inf_{\Psi(u) \leq B^2} L_{\mathcal{D}}(u) + (2\sqrt{2} + 1) \frac{GB}{\sqrt{\alpha n}}.$$

2.5 Problem 5. Soft-Margin Linear Classification

For hinge loss

$$\ell(w, (x, y)) = (1 - y\langle w, x \rangle)_+, \quad \|x\|_2 \leq R,$$

the map $w \mapsto \ell(w, (x, y))$ is R -Lipschitz in $\|\cdot\|_2$. Also

$$\Psi(w) = \frac{1}{2}\|w\|_2^2$$

is 1-strongly convex in $\|\cdot\|_2$.

To compete with all u satisfying

$$\|u\|_2 \leq B,$$

we have

$$\Psi(u) \leq \frac{B^2}{2}.$$

Using the optimized choice

$$\lambda = \sqrt{\frac{2R^2}{(B^2/2)n}} = \frac{2R}{B\sqrt{n}},$$

Problem 3 gives

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq L_{\mathcal{D}}(u) + \frac{2RB}{\sqrt{n}} + R\sqrt{\frac{2\eta}{\lambda}}.$$

Substituting $\lambda = 2R/(B\sqrt{n})$ into the optimization term,

$$R\sqrt{\frac{2\eta}{\lambda}} = R\sqrt{\frac{\eta B\sqrt{n}}{R}}.$$

Thus the approximate soft-margin RERM bound is

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq \inf_{\|u\|_2 \leq B} L_{\mathcal{D}}(u) + \frac{2RB}{\sqrt{n}} + R\sqrt{\frac{\eta B\sqrt{n}}{R}}.$$

To make the optimization-error term no larger than

$$\frac{RB}{\sqrt{n}},$$

it suffices that

$$R\sqrt{\frac{2\eta}{\lambda}} \leq \frac{RB}{\sqrt{n}}.$$

Equivalently,

$$\eta \leq \frac{\lambda B^2}{2n} = \frac{RB}{n^{3/2}}.$$

Under this condition,

$$\mathbb{E}_S L_{\mathcal{D}}(\tilde{w}_S) \leq \inf_{\|u\|_2 \leq B} L_{\mathcal{D}}(u) + \frac{3RB}{\sqrt{n}}.$$

3 Part C: A Weighted Euclidean Geometry

3.1 Problem 1. Geometry Quantities

Let

$$Q = \text{diag}(q_1, \dots, q_d), \quad q_j > 0.$$

The Q -norm is

$$\|w\|_Q = \sqrt{w^\top Q w}.$$

Its dual norm is

$$\|v\|_{Q,*} = \sqrt{v^\top Q^{-1} v}.$$

Indeed, by Cauchy–Schwarz,

$$\langle w, v \rangle = \langle Q^{1/2} w, Q^{-1/2} v \rangle \leq \|w\|_Q \sqrt{v^\top Q^{-1} v},$$

and equality is attained in the direction $w \propto Q^{-1} v$.

Therefore, using $|\phi_j(x)| \leq r_j$,

$$\|\phi(x)\|_{Q,*}^2 = \sum_{j=1}^d \frac{\phi_j(x)^2}{q_j} \leq \sum_{j=1}^d \frac{r_j^2}{q_j}.$$

Next,

$$\Psi_Q(w) = \frac{1}{2} w^\top Q w = \frac{1}{2} \sum_{j=1}^d q_j w_j^2.$$

Over

$$\mathcal{C}_b = \{w \in \mathbb{R}^d : |w_j| \leq b_j \ \forall j\},$$

each term is maximized by $|w_j| = b_j$. Hence

$$\sup_{w \in \mathcal{C}_b} \Psi_Q(w) = \frac{1}{2} \sum_{j=1}^d q_j b_j^2.$$

3.2 Problem 2. Fixed Q

The scalar loss is g -Lipschitz in the prediction, so for any w, w' ,

$$|\ell(\langle w, \phi(x) \rangle, y) - \ell(\langle w', \phi(x) \rangle, y)| \leq g |\langle w - w', \phi(x) \rangle|.$$

Using the Q dual norm,

$$|\langle w - w', \phi(x) \rangle| \leq \|w - w'\|_Q \|\phi(x)\|_{Q,*}.$$

Therefore the loss is G_Q -Lipschitz with respect to $\|\cdot\|_Q$, where

$$G_Q \leq g \sqrt{\sum_{j=1}^d \frac{r_j^2}{q_j}}.$$

Also Ψ_Q is 1-strongly convex with respect to $\|\cdot\|_Q$, and by Problem 1,

$$\sup_{w \in \mathcal{C}_b} \Psi_Q(w) = \frac{1}{2} \sum_{j=1}^d q_j b_j^2.$$

Apply the Week 8 stable-RERM theorem over $\mathcal{C}_b$:

$$\mathbb{E} L_{\mathcal{D}}(A_\lambda(S)) \leq \inf_{w \in \mathcal{C}_b} L_{\mathcal{D}}(w) + \lambda \left(\frac{1}{2} \sum_{j=1}^d q_j b_j^2 \right) + \frac{2g^2 \sum_{j=1}^d r_j^2 / q_j}{\lambda n}.$$

Let

$$A_Q := \sum_{j=1}^d q_j b_j^2, \quad R_Q^2 := \sum_{j=1}^d \frac{r_j^2}{q_j}.$$

The λ -dependent expression is

$$\frac{\lambda A_Q}{2} + \frac{2g^2 R_Q^2}{\lambda n}.$$

Optimizing over λ gives

$$\lambda^* = \frac{2g R_Q}{\sqrt{A_Q n}},$$

and the optimized value is

$$\frac{2g}{\sqrt{n}} \sqrt{A_Q R_Q^2}.$$

Thus

$$\mathbb{E} L_{\mathcal{D}}(A_{\lambda^*}(S)) \leq \inf_{w \in \mathcal{C}_b} L_{\mathcal{D}}(w) + \frac{2g}{\sqrt{n}} \sqrt{\left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)}.$$

3.3 Problem 3. Best Diagonal Geometry

We need to minimize

$$\left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)$$

over $q_j > 0$. By Cauchy–Schwarz,

$$\left(\sum_{j=1}^d b_j r_j \right)^2 = \left(\sum_{j=1}^d (\sqrt{q_j} b_j) \left(\frac{r_j}{\sqrt{q_j}} \right) \right)^2 \leq \left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right).$$

Equality holds when the two vectors are proportional, i.e. when

$$\sqrt{q_j} b_j = c \frac{r_j}{\sqrt{q_j}}$$

for a constant $c > 0$. Equivalently,

$$q_j = \frac{c r_j}{b_j}.$$

Since any $c > 0$ gives the same product, such a choice is valid for all j .

Therefore the minimum possible product is

$$\left(\sum_{j=1}^d b_j r_j \right)^2.$$

Plugging this into the fixed- Q bound from Problem 2, the best diagonal-geometry excess term is

$$\frac{2g}{\sqrt{n}} \sum_{j=1}^d b_j r_j.$$

4 Part D: AI Usage Report

I used AI for the proof organization, algebra checking, and LaTeX polishing in this assignment. The parts where AI was most useful were the validation-selection proof in Part A, the optimization-error constants in Part B, and the Cauchy–Schwarz geometry optimization in Part C.

The suggestions I accepted were the finite-class validation proof using Hoeffding’s lemma, the discretization calculation

$$t + \frac{1}{t} \leq \frac{5}{2} \quad \text{for } t \in [1/2, 2],$$

and the diagonal-geometry substitution

$$q_j \propto \frac{r_j}{b_j}.$$

These suggestions were useful because they directly matched the constants and structure requested in the assignment.

I rejected or substantially modified two suggestions. First, I did not use a high-probability validation argument, because the problem asks for an expectation bound and the cleaner expectation proof gives the exact stated form. Second, I corrected the weighted-geometry term to scale as

$$\frac{2g}{\sqrt{n}} \sum_j b_j r_j,$$

which follows from the Week 8 theorem; a version with $2g\sqrt{n}$ would have the wrong statistical rate.

I verified the submitted work by checking each proof against the assignment statement and the Week 8 slides. In Part A, I checked the validation term and the factor-2 grid discretization constants. In Part B, I checked the distinction between approximate-stability, which has a $2G\sqrt{2\eta/(\lambda\alpha)}$ term, and the expected-risk bound, which only needs one such comparison to the exact RERM output. In Part C, I checked the dual norm formula and the equality condition in Cauchy–Schwarz. Finally, I rebuilt the PDF with `pdflatex` to catch LaTeX and formatting errors.

No new repo skill was necessary for this assignment. The five most recent workflow rules I relied on were:

- use the shared `ai4lt` Conda environment before doing homework work,
- compile with `pdflatex`,
- create a reusable statement file for each homework folder,
- keep generated or supporting artifacts inside the assignment-local directory,
- keep long code out of the report and reference relative paths when code is used.